Geometric Occupancy from Weighted One-Quantum Gain/Loss Histories

Relation to Bose-Einstein and Planck Forms

Matter-as-machine

Abstract

This note studies a nonnegative integer occupancy $n = 0, 1, 2, \ldots$ in a discrete system. Level n contains n quanta and has energy $E_n = n\varepsilon$, where ε is the energy of one quantum. An elementary event changes n by one, subject to the boundary $n \geq 0$. First, a combinatorial ensemble is defined in which every allowed gain/loss history of length L has weight q^L. Its normalized endpoint distribution is exactly geometric. Second, a continuous-time birth-death process is considered. If each of the n quanta has loss rate μ, the total loss rate is $n\mu$. This process has a geometric stationary distribution if and only if the additional gain-rate condition $\lambda_n = (n+1)\lambda$ is imposed; that condition is not implied by the boundary or loss rule. Averaging E_n over the combinatorially derived geometric law gives $\frac{\varepsilon}{\exp(x)-1}$ after defining $x = -\log r_{\text{path}}$. This is a change of variable, not a physical derivation of an exponential mechanism. No relation between the combinatorial parameter q and the continuous-time ratio $\frac{\lambda}{\mu}$ is derived; equating their geometric ratios is an explicit matching condition. With the additional thermal identification $x = \beta(\varepsilon - \mu_{\text{chem}})$, the same geometric law has the standard single-mode Bose-Einstein form; for photons $\mu_{\text{chem}} = 0$, giving the Planck denominator. These thermal identifications are not derived here.

1. Nonnegative occupancy and one-quantum events

The system has occupancy

$$n = 0, 1, 2, 3, \ldots.$$

Level n contains n quanta. Each quantum contributes a fixed energy ε, so

$$E_0 = 0, \quad E_n = n\varepsilon \quad (n \geq 1).$$

Every elementary event changes n by one:

$$n \to n+1 \quad \text{(gain)}, \qquad n \to n-1 \quad \text{(loss)}.$$

A loss is forbidden at $n = 0$. Only one quantum is gained or lost in a single event. The state $n = 0$ is the empty level, and $0 \to 1$ is an ordinary gain event from that level. No particle identity or creation/death history is assigned to these transitions.

There are two different ways to assign weights to such histories:

- by the number of events in a history; or
- by rates per unit physical time.

These descriptions answer different questions and are treated separately. The constant path weight q is not a transition rate. In particular, the combinatorial calculation does not imply a value for $\frac{\lambda}{\mu}$ in the continuous-time model. Any equality between the two resulting geometric ratios is a later matching condition.

2. Constant-weight ensemble of finite histories

Consider all finite sequences of $+1$ and -1 steps that start at 0, never go below 0, and end at n. Assign weight q^L to every history of length L, where

$$0 < q < \frac{1}{2}.$$

More precisely, let $\mathcal{H}$ be the set of all such finite histories, including the empty history. Define

$$Z_{\text{path}} = \sum_{h\in\mathcal{H}} q^{L(h)}, \qquad P(h) = \frac{q^{L(h)}}{Z_{\text{path}}}.$$

Since the number of length-L histories is at most 2^L, the condition $q < \frac{1}{2}$ guarantees that Z_{path} is finite. Thus q is a history-length weight, not a one-step transition probability, and no separate stopping rule is assumed.

If the endpoint is n, then $L = n + 2k$ for some $k \geq 0$. Such a history contains $n + k$ upward steps and k downward steps. The ballot theorem gives the number of allowed histories:

$$\frac{n+1}{n+k+1}\binom{n+2k}{k}.$$

Consequently, the total endpoint weight is

$$W(n) = \sum_{k=0}^{\infty} \frac{n+1}{n+k+1}\binom{n+2k}{k} q^{n+2k}.$$

2.1. Small endpoint examples

For endpoint $n = 1$, the numbers of allowed histories at lengths $1, 3, 5, 7, \ldots$ are $1, 2, 5, 14, \ldots$. Their total endpoint weight begins

$$W(1) = q + 2q^3 + 5q^5 + 14q^7 + \ldots.$$

For endpoint $n = 2$, the corresponding counts at lengths $2, 4, 6, 8, \ldots$ are $1, 3, 9, 28, \ldots$, giving

$$W(2) = q^2 + 3q^4 + 9q^6 + 28q^8 + \ldots.$$

These examples illustrate the competition in the full sum: the number of allowed histories grows with length, while the factor q^L suppresses longer histories. The next section evaluates all such sums exactly.

3. Exact evaluation of the endpoint weight

Let

$$C(z) = \frac{1-\sqrt{1-4z}}{2z}$$

be the generating function of the Catalan numbers. The generalized Catalan identity is

$$\sum_{k=0}^{\infty} \frac{n+1}{n+k+1}\binom{n+2k}{k} z^k = C(z)^{n+1}.$$

A proof is given in Appendix A. Setting $z = q^2$ gives

$$W(n) = q^n C(q^2)^{n+1}.$$

Define

$$r_{\text{path}} = qC(q^2) = \frac{1-\sqrt{1-4q^2}}{2q}.$$

The Catalan functional equation $C(z) = 1 + zC(z)^2$ implies

$$r_{\text{path}} = q\left(1 + r_{\text{path}}^2\right),$$

or equivalently

$$qr_{\text{path}}^2 - r_{\text{path}} + q = 0.$$

Solving for q gives

$$q = \frac{r_{\text{path}}}{1 + r_{\text{path}}^2}.$$

For every $0 < r_{\text{path}} < 1$, this gives $q > 0$ and $q < \frac{1}{2}$, because

$$2r_{\text{path}} < 1 + r_{\text{path}}^2 \quad \text{if and only if} \quad \left(1 - r_{\text{path}}\right)^2 > 0.$$

Moreover,

$$\frac{\mathrm{d}q}{\mathrm{d}r_{\text{path}}} = \frac{1 - r_{\text{path}}^2}{\left(1 + r_{\text{path}}^2\right)^2} > 0.$$

The map approaches 0 at the lower endpoint and $\frac{1}{2}$ at the upper endpoint. Hence it is a bijection from $0 < r_{\text{path}} < 1$ onto $0 < q < \frac{1}{2}$; the path construction realizes the entire normalizable geometric family.

It follows that

$$W(n) = \left(\frac{r_{\text{path}}}{q}\right) r_{\text{path}}^{n} = \left(1 + r_{\text{path}}^{2}\right) r_{\text{path}}^{n}.$$

Normalizing over all $n \geq 0$ gives

$$P_{\text{path}(n)} = \frac{W(n)}{\sum_{m=0}^{\infty} W(m)} = (1 - r_{\text{path}}) r_{\text{path}}^{n}.$$

Thus the constant-weight ensemble of finite histories has an exact geometric endpoint occupancy distribution. This is a combinatorial statement: q weights complete event sequences and is not, by itself, a rate per unit time.

4. Continuous-time gain/loss process

Suppose each of the n quanta has the same independent loss rate μ. Although only one quantum leaves in any individual event, the total loss rate is

$$\mu_n = n\mu.$$

For $n \geq 1$, the mean waiting time until the next loss, if no competing gain occurs, is $\frac{1}{n\mu}$. At $n = 0$, the loss rate is zero.

Let λ_n be the gain rate at occupancy n. The continuous-time birth-death chain has transitions

$$n \to n+1 \quad \text{at rate} \quad \lambda_n, \qquad n \to n-1 \quad \text{at rate} \quad n\mu.$$

If a stationary distribution $\pi(n)$ exists and the stationary probability current vanishes, detailed balance gives

$$\pi(n)\lambda_n = \pi(n+1)(n+1)\mu,$$

and therefore

$$\frac{\pi(n+1)}{\pi(n)} = \frac{\lambda_n}{(n+1)\mu}.$$

Iteration gives the formal stationary weights

$$\pi(n) = \pi(0) \prod_{j=0}^{n-1} \frac{\lambda_j}{(j+1)\mu}, \qquad n \geq 1.$$

Define the normalization series

$$Z = 1 + \sum_{n=1}^{\infty} \prod_{j=0}^{n-1} \frac{\lambda_j}{(j+1)\mu}.$$

Assuming the chain is well defined and nonexplosive, a normalized stationary distribution exists precisely when $Z < \infty$. In that case $\pi(0) = \frac{1}{Z}$ and the product formula determines every $\pi(n)$. If Z diverges, the detailed-balance weights cannot be normalized.

5. Condition for a geometric stationary distribution

A geometric stationary law requires $\frac{\pi(n+1)}{\pi(n)}$ to be a constant r_{rate} with $0 < r_{\text{rate}} < 1$. Therefore the necessary and sufficient gain-rate condition is

$$\lambda_n = (n+1)\lambda, \qquad r_{\text{rate}} = \frac{\lambda}{\mu} < 1.$$

This is a condition for geometricity, not a derivation of the gain law. The boundary permits $\lambda_0 > 0$, and independent per-quantum loss specifies the downward rate $n\mu$, but neither assumption determines λ_n for $n \geq 1$. Choosing λ_n proportional to $n+1$ makes the stationary ratio constant by construction. A physical use of this model therefore requires a separate microscopic reason for the factor $n+1$.

Under this condition,

$$\pi(n+1) = r_{\text{rate}}\pi(n), \qquad \pi(n) = (1 - r_{\text{rate}}) r_{\text{rate}}^{n}.$$

At $n = 0$, the rate $\lambda_0 = \lambda$ is the gain rate from the empty level. Independent per-quantum loss explains why one quantum is lost per event while the total loss frequency grows as n. It does not explain the geometric law.

For comparison, if $\lambda_n = \lambda$ is constant, then

$$\frac{\pi(n+1)}{\pi(n)} = \frac{\lambda}{(n+1)\mu},$$

and the normalized stationary law is Poisson:

$$\pi(n) = \exp\left(-\frac{\lambda}{\mu}\right)\frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}.$$

Thus n-dependent loss alone does not imply a geometric distribution.

6. Mean occupancy, mean energy, and thermal correspondence

Assign level energy $E_n = n\varepsilon$, with $E_0 = 0$. For a normalized geometric law

$$P(n) = (1-\rho)\rho^n, \qquad 0 < \rho < 1,$$

where ρ may be r_{path} or r_{rate} under their respective assumptions,

$$\langle n\rangle = \sum_{n=0}^{\infty} n(1-\rho)\rho^n = \frac{\rho}{1-\rho}.$$

Writing $\rho = \exp(-x)$, where $x > 0$, yields

$$\langle n\rangle = \frac{1}{\exp(x)-1}, \qquad \langle E\rangle = \frac{\varepsilon}{\exp(x)-1}.$$

6.1. Relation to Bose-Einstein and Planck forms

The derived geometric law has the same mathematical form as the occupation distribution of one bosonic mode. If one additionally identifies

$$x = \beta(\varepsilon - \mu_{\text{chem}}), \qquad \beta = \frac{1}{kT},$$

normalizability requires $\varepsilon - \mu_{\text{chem}} > 0$. Substitution gives the full single-mode Bose-Einstein probability law

$$P(n) = [1 - \exp(-\beta(\varepsilon - \mu_{\text{chem}}))]\exp(-n\beta(\varepsilon - \mu_{\text{chem}})),$$

and its mean

$$\langle n\rangle = \frac{1}{\exp(\beta(\varepsilon - \mu_{\text{chem}})) - 1}.$$

For photons in thermal equilibrium, setting $\mu_{\text{chem}} = 0$ gives

$$\langle E\rangle = \frac{\varepsilon}{\exp(\frac{\varepsilon}{kT}) - 1},$$

the Planck mean-energy denominator for one mode. The present note derives the geometric occupancy law. It does not derive the quantum-mechanical reason that physical bosons obey this law, the thermal origin of x, or the zero photon chemical potential. Here $x = -\log\rho$ is only a parametrization of the geometric ratio. A physical interpretation of x requires an additional model.

7. Relation between the two descriptions

Both constructions produce a geometric formula under their respective assumptions, but for different reasons. In the constant-weight path ensemble, geometricity follows by summing Catalan-counted histories with weight q^L; physical time and state-dependent waiting times are absent. In the continuous-time process, geometricity is a stationary consequence of

$$\frac{\lambda_n}{(n+1)\mu} = r_{\text{rate}} = \frac{\lambda}{\mu}.$$

The two parameters are

$$r_{\text{path}} = \frac{1-\sqrt{1-4q^2}}{2q}, \qquad r_{\text{rate}} = \frac{\lambda}{\mu}.$$

They should not be identified without an additional model connecting path-length weights to physical transition rates. If both descriptions are required to represent the same law, one may impose $r_{\text{path}} = r_{\text{rate}}$, but this is a matching condition, not a consequence of either construction.

8. Relation to cited prior results

The two mathematical ingredients used here have established counterparts in different literatures. The endpoint counts in Sections 2-3 are classical ballot numbers belonging to the Catalan triangle, and powers of the Catalan generating function are standard combinatorial objects [1]. The continuous-time process in Sections 4-5 is a special case of the linear birth-death process with immigration studied by Giorno and Nobile [2], whose rates are

$$\lambda_n = n\lambda + \nu, \qquad \mu_n = n\mu.$$

Setting the immigration rate $\nu = \lambda$ gives exactly $\lambda_n = (n+1)\lambda$ and $\mu_n = n\mu$. In the subcritical case $\lambda < \mu$, their negative-binomial steady state reduces to the geometric law when $\frac{\nu}{\lambda} = 1$.

In the cited Catalan-triangle treatment, the identities are studied as combinatorial sums, without the occupancy interpretation used here. In the cited birth-death treatment, the distribution is obtained from a continuous-time rate model. The present note instead places beside those results a finite-history ensemble in which every complete nonnegative path has weight q^L. This is the specific construction and comparison offered here. No broader claim is made about all Catalan-path or master-equation literature.

9. Conclusion

The weighted-history calculation gives an exact combinatorial construction of a geometric occupancy distribution on $n = 0, 1, 2, \ldots$. Level n contains n quanta with energy $E_n = n\varepsilon$, and $n = 0$ is the empty level. Averaging over the distribution gives

$$\langle E \rangle = \frac{\varepsilon}{\exp(x) - 1}.$$

Because $q = \frac{r_{\text{path}}}{1+r_{\text{path}}^2}$ maps $0 < r_{\text{path}} < 1$ bijectively onto $0 < q < \frac{1}{2}$, the construction realizes the full normalizable geometric family. After the additional identification $x = \beta(\varepsilon - \mu_{\text{chem}})$, the law can be written in the standard single-mode Bose-Einstein form; for photons, $\mu_{\text{chem}} = 0$ gives the corresponding Planck denominator. These are optional physical interpretations, not derivations of the quantum-mechanical or thermal origin of the law.

The continuous-time construction is separate. Per-quantum loss gives total loss rate $n\mu$, but the boundary and loss rules alone do not produce a geometric stationary law. That law occurs only when $\lambda_n = (n+1)\lambda$ is imposed. A physical origin for this gain law and for the thermal meaning of x remains additional work.

Appendix A: Proof of the generalized Catalan identity

Let $C(z) = 1 + zC(z)^2$ and define $U(z) = C(z) - 1$. Then

$$U = z(1+U)^2.$$

We seek the coefficient of z^k in $C(z)^{n+1} = (1+U(z))^{n+1}$. For $k \geq 1$, the Lagrange inversion formula gives

$$[z^k](1+U(z))^{n+1} = \frac{1}{k}[u^{k-1}](n+1)(1+u)^{n(1+u)^{2k}} = \frac{n+1}{k}\binom{n+2k}{k-1}.$$

The binomial identity

$$\frac{n+1}{k}\binom{n+2k}{k-1} = \frac{n+1}{n+k+1}\binom{n+2k}{k}$$

therefore implies

$$[z^k]C(z)^{n+1} = \frac{n+1}{n+k+1}\binom{n+2k}{k}.$$

For $k = 0$, both sides equal 1. Hence

$$C(z)^{n+1} = \sum_{k=0}^{\infty} \frac{n+1}{n+k+1}\binom{n+2k}{k} z^k,$$

which is the identity used in Section 3.

References

[1] Y. Sun and F. Ma, "Some New Binomial Sums Related to the Catalan Triangle," *The Electronic Journal of Combinatorics* **21**(1) (2014), Article P1.33. DOI: 10.37236/3701.

[2] V. Giorno and A. G. Nobile, "Bell Polynomial Approach for Time-Inhomogeneous Linear Birth-Death Process with Immigration," *Mathematics* **8**(7) (2020), Article 1123; see Section 2.3, especially equations (20)-(24). DOI: 10.3390/math8071123.